BIO 301 (Essential of Genetics)

Final Term current File 2023

In Association: Work Together Team

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----------Objective----------

1. **Female bee sex is_____haploid**
2. **Recovery of DNA is called_____ DNA extraction or DNA isolation**
3. **Autosomal recessive _____cystic fibrosis**
4. **Mendel's work_____ 3 laws**
5. **X0 system in_____insects**
6. **Xo trait s present in_____fruitfly**
7. **Nature has variations ____ widespread**
8. **"m." For a mitochondrial sequence _____(like m.8993T>C,)**
9. **Rabbit having light grey colour_____chinchila**
10. **Polygenic traits are----(qualitative)**
11. **In Giemsa staining Dark region bands are----- and light region bands are----- (A-T and G-C).**
12. **Arm of the chromosomes are named as------ (p and q)**

13. **Genetically determined variability among humans is due to a difference in---- of the genomic sequence. (0.1%)**

14. **In TELOPHASE I------set of chromosomes arrived at each of the pole.___(Haploid)**

15. **…..is used for identifying of SNPs__ Genotyping**

16. **Annealing temperature….. 40-65°C**

17. **in R banding darker region of chromosome represent…… Euchromatin**

18. **origin of …….type of mutation unknown__ spontaneous mutation**

19. **cri- du- chat syndrom is caused by …Mutation.**

20. **monosomy represent as ….. base___ 2n-1**

21. **lampbrush chromosome are found in… 1882**

22. **Fragile X syndrome is …. X-linked dominant disease**

23. **Epistasis is an interaction of how many genes.. two or more**

24. **HGNC stands for… (HUGO gene Nomenclature committee)**

25. **who discovered genetics .. Gregor Mendel**

26. **Two Barr bodies are due to deactivation of two____ X chromosomes**

27. **Which is not Autosomal dominant disease? __Gaucher's disease**

28. **In ZW-ZZ system, which is male? ___ZZ is male**

29. **Down syndrome is due to trisomy of ___chromosome 21.**

W.T.T

30. _____is non-ionic detergent, designed to lyse outer cell membrane of Blood and epithelial cells, but will not break down nuclear membrane. ___Cell Lysis Buffer

31. Darker bands in R staining are ___Euchromatin.

32. Annealing temperature in PCR ___40-65°C.

33. ____ is commonly used to determine whether a DNA sequence is present in a sample and number of its copies in the sample___ Q-PCR

34. ___used for identifying of SNPs.__ allele specific PCR

35. Whole genomes can be amplified by ___Whole genome PCR.

36. Mini Primer PCR can extend from ____short primers.

37. Northern blot is a method used for detection of a specific ____sequence.__ RNA

38. Which describes best the prokaryotic chromosome ____ Single, circular DNA molecule.

39. Prokaryotic nucleus lies in ___nucleoid.

40. ____are used to determine the pattern of inheritance of a genetic disease within a family.__ Pedigrees

41. Variations are due to ___Genes & environment.

42. In ring chromosome, two ___ joined joins to form ring shape. ___ arms

43. Human zygote is ___ chromosomes. Diploid and 46

44. ___ Histone tightly pack the nucleosome.__H1

45. Z-DNA is ____ helix. __Left-handed

W.T.T

46. **Which is Transition mutation?___ Purine is replaced by purine.Or Pyrimidine is replaced by pyrimidine.____ Both are correct.**

47. **Growth takes place by ____Mitosis.**

48. **____push against each other, that stretch the cell into an oval shape in Anaphase. ___ Non-kinetochore spindle fibers**

49. **Meiosis I results in two ___ daughter cells. __ haploid**

50. **Cystic fibrosis and muscular dystrophy are ___Single gene disorders.**

51. **Genes switched on when have ___ cytosines.__unmethylated**

52. **Trisomy is ___ Aneuploidy.**

53. **____produces a change in amino acid sequence in protein product.___ Missense mutation**

54. **Which is not true about Autosomal recessive disease?____ Diseases occurs in individuals with two mutant alleles. This one is not true.**

55. **Homozygotes with two copies of the altered gene. It's true.**

56. **Frequency of ___ is highest. ___ autosomal dominant disorders**

57. **Microbiology, DNA fingerprinting, Genetic diversity, Forensics, Mapping of genomes are applications of?___VNTRs.**

58. **Male pattern baldness is a recessive sex-linked trait on the X chromosome. A woman, whose father had male pattern baldness, marries a man with this trait. What is the probability that any son born will have pattern baldness?**

    The probability that a son will be born with pattern baldness is 50%.

59. **Both a husband and wife have normal vision but their son is color blind. What are the likely genotypes of the boy's parents?**

    Boy = XaY (recessive from mother = Colorblind son)

60. Blood cells are lysed ___erythrocytes lysis and leukocytes lysis.

61. Dissolve RNA pellet by adding ___RNAse-free H2O or buffer

62. In PCR, DNA is ___ amplified.

63. Which is an exception in Mendelian inheritance ___multiple alleles

64. Trisomy 18 causes which disease ___Edward syndrome

65. Who discovered gene map in Drosophilla?__T.H Morgan

66. How many no.of genes are present in human mitochondrial genome __37.

67. Which of the following is not an example of multi-locus marker ___SST

68. Which PCR is used to determine SNPs ___ Allelic specific PCR

69. Dark bands are rich in __A.T light bands are rich in ___G.C

70. Whole genomic size & amplification is done by ___Whole genome PCR

71. Fur color in Labrador retriever controlled by ___2 genes.

72. Annealing temp__ 40-65°C

73. ___ is used to identify region by using probes.__Southern blotting

74. Northern blotting is used to for ___ RNA

75. Okazaki fragments occur in ___discontinuous replication

The cross of PPTT and pptt, F1 dominant progeny percentage is___100%.

76. 🚫 Symbol means___ dead

77. Which is not related to X-linked trait…? it is present in females more than male

78. Chromosome number in round worm__ 2

79. In which stage of Meiosis I chormosomes are replicated..? __Anaphase

80. In which phase of meiosis, chormosomes are replicated. (Options meiosis I, meiosis II, ___Both

81. Chromosome can be counted only in ….. Phase..? __Metaphase

82. ___Can change color of hydrogenae flower.__ PH of soil

83. Which part of chromosome is visible when viewed under microscope… chromatid

84. ___cases of turner syndrome are due to ring chromosome.___ 20%

85. Annealing temperature in PCR ..40- 65°C

86. Which blood group is universal recipient___ AB blood group

87. About 25000 SNA in human are _____ non- synonymous

88. A condition in which some people with an appropriate genotype fail to express the phenotupe is called as ____ penetrance

89. Pattern baldness is a recessive trait. A female inherits it from her father and marries a man who has this trait . What are the chances to pass on this trait 50%

90. Expression of phenotype by a number of individuals is called as_____ phenotypic variation

91. Geimsa stain is used for ___ chromosomal banding (C-banding)

92. Northern blotting is used for____ RNA

93. Southern blot is use for ____ DNA

W.T.T

94. Cri- du- chat syndrom is caused by … mutation.

95. The Type of cell division important for growth is___ mitosis

96. R banding has dark bands of____ euchromatic (G-C)

97. cancer, infertility are result of _____ Multifactorial disease

98. Colchicine has the extremely useful property of preventing formation of the ___Spindle,

99. Autosomal dominant diseases____ Huntington's disease

100. ____is inherited through ovum not sperm__ Mitochondrial DNA

101. Standard present in at least 1–2% of a population_____ DNA polymorphism

102. A smooth curve is property of …Continuous variation

103. DNA replication is …. Semi conservative

104. In case of X linked recessive disorder A Cross b/w abnormal and normal female Will be … 50%

105. Multifactorial disease caused by….. genes and environmental effect

106. Addition mutation is the addition of….. nucleotides/base pairs

107. ……is a chelating agent.__ EDTA

108. PCR is a technique used for nucleic acid… Amplification

109. Most common inversion seen in human is on chromosomes……..at inv p12 q13__ nine

110. The disease is said to have ….If all individual who have disease caused mutation have clinical symptoms . Complete penetrance

111. **For high resolution chromosomes analysis which stages of cell cycle are cell arrested before staining….. metaphase stage**

112. **Chromosomes……lead to down syndrome. Chromosome 21**

113. **Monosomy represent as ……. Base . 2n-1.**

114. **If 14 % nucleotide from DNA molecule contains base T what present will contain base G ??.... 36%**

115. **Cell splits into two in …… mitosis**

116. **Types of cell division important for growth is … cytokinesis or telophase**

117. **Haploid gametes are formed by ….. meiosis**

118. **How many numbers of alleles control of color of rabbit …… four**

119. **It is estimate that about ……Genes are imprinted can found on several different chromosomes. 40**

120. **CML stand for …. Myelogenous leukemia**

121. **Which type of mutation involved in significantly reduce the longivty of individual…. Lethal mutation**

122. **Which is example of X linked disorder…. Hunter's disease**

123. **In pedegre analysis symbol ⃠ represent…… dead**

124. **Which is not example of multi locus markes…. STS**

125. **Mutation can be identify by ……blotting**

126. **Chromosomes have centromere at one end of the arms … telocentric**

127. **Ants and bees have …. No sex chromosome**

W.T.T

128. About 25000 SNPs in humans are … non- synonymous

129. Lamp brush chromosomes were discovered in …. 1882

130. The idea that different pairs of alleles are passed to offspring independently Mendel principal of …. Law of independent assortment

131. Mendelian inheritance the gene that obey ….. principal o segregation and independent assortment

132. Skin color of pigment in humans is controlled by at least …. Separately inherited genes . Three

133. ……banding pattern of chromosomes used variety of technique at stain tha GC Rich DNA on the chromosomes.__ Giemsa staining

134. The nucleosome is composed of a score of ……. Histone proteins and DNA ……BP wrapped around them. ___8, and 147

135. How many types of histone molecules in nature ?……. Eight

136. Which can be seen using the microscope during karyotyping……. Chromosomes

137. More than two sets of chromosomes which are coming from different parents…….Polyploidy

138. Monosomy represent as…. 2n-1

139. During…..of DNA the coiling must be released.__ transcription

140. During recessive epistasis …gene exerts its effect with…. Genotype .__ one, homozygous recessive

141. How many cases of Turner syndrome are due to using chromosomes … 20%

142. Which type of mutation occurs boundaries of exons and intron … splice mutation

143. **Which disease is due to deletion in chromosomes … Cri-du-chat syndrome**

144. **An example of autosomal dominant disorder is …..Huntington's disease**

145. **During DNA purification RNA pellets is dissolved by adding……to each sample__ RNase free water or buffer**

146. **In organic method of DNA purification….is used to remove the protein.__ phenol chloroform**

147. **Whole genome application PCR can be used to amplify … whole Genome**

148. **…..is used for identify SNPs .__ genotyping**

149. **Method for DNA purification include … (organic, inorganic, chelex, FTA, silica based)**

150. **Blood cells are type in …. Blood typing Procedure .**

151. **Prokaryotic nucleus lies in __nucleoid.**

152. **Unit of genetic map……. Centimorgan**

153. **Who discover Restrictions Endonuclease…… Arber, Smith&Nathans**

154. **Protein activate after dna destroy……. Histone**

155. **DNA Replicate at which stage of interphase……. S phase**

156. **Division of cytoplasm called…… Cytokinesis**

157. **An average cell can divide ……. Before it Gets unhealthy__. 50 Times**

158. **The unspecialized cell that renew …… Stem Cells**

159. **Bones regenerate every…… 10 years**

160. **Golgi apparatus found in…… Eukaryotic Cells**

W.T.T

161. **T-cells activate …… B-cells**

162. **A cross for determination of the genotype of an individual is called a_____**
test cross

163. **Deformities in pregnant women is caused by drug____thalidomide**

----------Subjective----------

Deletion mutation and it's cause and effect

Deletion of one or more nucleotides in DNA sequence is called deletion mutations.

Deletion mutations can occur due to various factors such as errors during DNA replication,exposure to mutagenic substances, or spontaneous changes in DNA.

Missense mutations and its effect on protein formation

Missense mutation: Missense mutations substitute one amino acid for another. Some missense mutations have very large effects. Some have minimal or no effect. It depends onwhere the mutation occurs in the protein's structure and how big a change in the type ofamino acid it is. Missense mutation produces a change in amino acid sequence in protein product (e.g. Histidine in for Arginine). May change function of protein or may not

Comparison between continuous and discontinuous variations

Continuous variation refers to traits that can take on a range of values within a population, such as height or weight. These traits are influenced by multiple genes and environmental factors, resulting in a wide spectrum of possible values.

Discontinuous variation, on the other hand, refers to traits that can be categorized into distinct groups or phenotypes, such as blood type or eye color. These traits are usually controlled by a small number of genes with specific alleles, leading to clear-cut differences between categories.

Erythrocytes lysis in RNA purification

In RNA purification, erythrocyte lysis is a step where red blood cells are broken open to release their contents. This is usually done using a lysis buffer that disrupts the cell membranes and releases the RNA. The lysate is then processed further to purify the RNA of interest.

Procedure of southern blotting

1. DNA Digestion: The DNA sample is cut into fragments using restriction enzymes.
2. Gel Electrophoresis: Fragments are separated by size using gel electrophoresis.
3. Denaturation: DNA fragments are treated to separate the double-stranded DNA into single strands.
4. Transfer: Fragments are transferred from the gel to a membrane, like nitrocellulose or nylon.
5. Hybridization: A labeled DNA probe binds to the target DNA sequences on the membrane.
6. Washing: Non-specifically bound probes are washed away, leaving only the hybridized ones.
7. Detection: The presence of the labeled probe is visualized using autoradiography or other detection methods.

This technique is used to detect specific DNA sequences and analyze DNA fragment sizes.

Complications of inheritance…

There are five basic modes of inheritance for single-gene diseases:

1. autosomal dominant,

2. autosomal recessive,

3. X-linked dominant,

4. X-linked recessive,

5. and mitochondrial.

X- linked diseases

X-linked recessive: Mutated alleles are present on either of the sex chromosomes.

X-linked dominant: inheritance refers to genetic conditions associated with mutations in genes on the X chromosome.

Families with an X-linked recessive disorder often have affected males, but rarely affected females, in each generation. For X-linked dominant diseases, however, a mutation in one copy of an X-linked gene will result in disease for both males and females.

Quantitation of RNA.

Quantification of RNA involves measuring the amount of RNA molecules in a sample. Common methods include spectrophotometry, qRT-PCR, and RNA sequencing. The chosen method depends on the purpose and sensitivity required for the analysis.

What are sex chromosomes, What are homogametic and heterogametic

Sex chromosomes are a pair of chromosomes that determine an individual's sex. In humans, they are typically denoted as X and Y. Males have one X and one Y chromosome (XY), while females have two X chromosomes (XX).

Homogametic refers to the sex that has two identical sex chromosomes, either XX (in females) or YY (in certain male animals). **Heterogametic** refers to the sex that has two different sex chromosomes, typically XY (in males). In humans, females are homogametic (XX) and males are heterogametic (XY).

Spontaneous Mutation and mutagenic agent

Spontaneous Mutation:

A spontaneous mutation is a genetic mutation that arises naturally in the absence of any external factors or influences. These mutations occur due to errors in DNA replication, repair, or recombination processes that are part of the normal cellular machinery.

Spontaneous mutations can be caused by intrinsic factors such as the inherent instability of certain DNA sequences or by the occasional mispairing of nucleotides during DNA synthesis. They can lead to changes in the DNA sequence, which can have various effects on gene function and protein production.

Mutagens:

Mutagens are agents that increase the rate of mutation above the spontaneous mutation rate. They can induce changes in the DNA sequence by causing chemical modifications, DNA strand breaks, or other forms of damage. Mutagens can be classified into two main categories:

Physical Mutagens: These mutagens include physical agents that directly interact with DNA and cause damage. Examples of physical mutagens include ionizing radiation (such as X-rays and gamma rays) and ultraviolet (UV) radiation. These agents can lead to the formation of DNA lesions and errors during DNA repair.

Chemical Mutagens: Chemical mutagens are compounds that interact with DNA and cause chemical modifications. These modifications can lead to base substitutions, insertions, deletions, or other structural changes in the DNA sequence. Examples of chemical mutagens include certain chemicals found in the environment, tobacco smoke, industrial pollutants, and some cancer-causing agents.

Causes of polyploidy

Polyploidy is a genetic condition where an organism has more than two sets of chromosomes in its cells. It can occur naturally or be induced artificially.

Natural causes include errors in cell division, resulting in the duplication of chromosomes, while artificial causes involve interventions like chemicals or irradiation to induce extra chromosome sets. Polyploidy can lead to increased genetic diversity, altered physical traits, and sometimes even enhanced adaptability to different environments.

Giemsa Stain

Geimsa stain is a type of stain used in microscopy to visualize and differentiate different types of cells, especially blood cells. It works by staining the DNA and RNA in the cells, allowing for better visualization of cellular structures and identification of specific cell types.

Hot Start PCR

Hot start technique is based on the principle of initiating the polymerase reaction at or above the primer annealing temperature, thereby preempting possible extensions of the primers nonspecifically annealed at ambient temperature (20°-55°C) at which Taq Pol retains partial activity. For some applications, hot-start PCR may be the only way to amplify successfully the desired product(s) with high yield and specificity. Hot Start PCR allows for reaction set up at room temperature without non-specific amplification and primer
dimer formation

Difference between inorganic and organic method of DNA Extraction and write steps of organic method of DNA Extraction

The main difference between the inorganic method and the organic method of DNA extraction lies in the type of reagents used. In the inorganic method, substances like salts and alcohols are used to precipitate DNA, while the organic method involves the use of organic solvents like phenol and chloroform.

Difference between oogenesis and spermatogenesis

Spermatogenesis is the process of sperm cell development in males, which occurs in the testes. It involves the production of mature sperm cells through a series of cell divisions and differentiation.

On the other hand, oogenesis is the process of egg cell development in females, which occurs in the ovaries. It involves the production of mature egg cells through a series of cell divisions, with

only one functional egg cell being produced from each cycle. Both spermatogenesis and oogenesis are essential for sexual reproduction and the formation of offspring

Role of mitosis and meiosis in animal

Mitosis and meiosis are two different processes involved in cell division.

Mitosis:

- Occurs in somatic (body) cells.

- The purpose is to produce two identical daughter cells with the same number of chromosomes as the parent cell.

- Consists of one round of cell division, resulting in two diploid (2n) daughter cells.

- Each daughter cell has the same genetic information as the parent cell.

- Plays a role in growth, repair, and asexual reproduction.

Meiosis:

- Occurs in cells of the reproductive organs (gonads) to produce gametes (sperm and eggs).

- The purpose is to reduce the chromosome number by half and introduce genetic variation.

- Consists of two rounds of cell division, resulting in four haploid (n) daughter cells.

- Each daughter cell has half the number of chromosomes as the parent cell.

- Plays a role in sexual reproduction and ensures genetic diversity.

How can we identify chromosomes

We can identify chromosomes through various techniques, such as karyotyping, fluorescent in situ hybridization (FISH), or chromosomal microarray analysis (CMA). These methods allow us to visualize and analyze the structure and number of chromosomes in a cell.

W.T.T

Difference between hot star and suicide PCR

“Hot start” PCR is a technique used to prevent non-specific amplification during the initial steps of the polymerase chain reaction (PCR). This is achieved by adding a DNA polymerase inhibitor to the reaction mix that becomes inactive at lower temperatures. This helps to reduce background noise and improve specificity in PCR reactions.

Suicide PCR method prescribes the use of any primer combination only once in a PCR (hence the term “suicide”), which should never have been used in any positive control PCR reaction, And the primers should always target a genomic region never amplified before in the lab using this or any other set of primers

Eye color, hair, facial features, appendages write recessive and dominant

here’s a breakdown of which traits are typically considered dominant and recessive:

1. Eye Color

- Dominant: Brown

- Recessive: Blue

2. Hair Color:

- Dominant: Dark colors (Black, Brown)

- Recessive: Light colors (Blonde, Red)

3.Facial Features** (Dimples as an example):

- Dominant: Dimples

- Recessive: No dimples

4. Appendages (Freckles on arms as an example):

- Dominant: Freckles on arms

W.T.T

- Recessive: No freckles on arms

Plants examples triploid tetraploid hexaploid octaploid and decaploid

here are examples of ploidy levels in plants:

1. **Triploid**: A plant with three sets of chromosomes.
   Example: Seedless watermelons are often triploid. They are created by crossing a diploid watermelon with a tetraploid watermelon.
2. **Tetraploid**: A plant with four sets of chromosomes.
   Example: Durum wheat (Triticum turgidum) is often cultivated in a tetraploid form.
3. **Hexaploid**: A plant with six sets of chromosomes.
   Example: Bread wheat (Triticum aestivum) is a hexaploid. It's one of the most widely cultivated crops for its edible grains.
4. **Octaploid**: A plant with eight sets of chromosomes.
   Example: The cultivated strawberry, Fragaria x ananassa, is often octaploid.
5. **Decaploid**: A plant with ten sets of chromosomes.
   Example: Some varieties of sugar cane can be decaploid.

Short range applications of molecular markers

Parentage determination

Genetic distance estimation

Determination of twin zygosity

Sexing of pre-implanted embryos

Identification of disease carriers

Percentage of mono,di, poly allelic in human

 30% genes in humans are di-allelic they exist in two forms.

 70% are mono-allelic, they exist only in one form.

 Few are poly-allelic.

Linked and causative single polymorphism.

Causative SNPs in gene:

Non coding SNP:change amount of protein produced. Coding SNP change amino acid sequence Linked SNPs, or single nucleotide polymorphisms, are genetic variations that are located close together on a chromosome. These SNPs tend to be inherited together as a block, rather than independently.. Linked SNPs can be used in genetic studies to identify associations with certain traits or diseases.

Polyploidy and it's types

Polyploidy is the heritable condition of possessing more than two complete sets of chromosomes.

Multiples of haploid cells or no of chromosomes are called polyploidy. It can be in form of triploidy, tetraploidy , pentaploidy etc. during polyploidy fetus does not survive this phenomena is seems in many crops.

Two major types

1.**Autopolyploidy**

Autopolyploids are polyploids with multiple chromosome sets dervied from a single species.

2 . **Allopolyploidy**

in case of allopolyploids are polypoloids with chromosomes derived from different species in the result of cross between two different species. Interspecies crosses can generate alloploids and their offspring generally sterile.

Genetic/inherited variation and it's two examples

Genetic or inherited variations refer to the differences in genetic information that are passed down from parents to their offspring. These variations can occur due to changes in DNA sequences, such as mutations or genetic recombination during the formation of reproductive cells.

Some examples of genetic variations include:

1. Eye color 2. Blood type 3. Height: 4. Skin color 5. Inherited diseases

Principle of PCR

The PCR technique is based on the enzymatic replication of DNA. In PCR, a short segment of DNA is amplified using primer mediated enzymes. DNA Polymerase synthesises new strands of DNA complementary to the template DNA. The DNA polymerase can add a nucleotide to the pre-existing 3'-OH group only

Procedure for western blotting

The Western blotting procedure involves several steps. First, proteins are separated by size using gel electrophoresis. Then, the proteins are transferred onto a membrane. The membrane is blocked to prevent non-specific binding, and then incubated with a primary antibody that binds to the target protein. After washing, a secondary antibody is added that recognizes the primary antibody. Finally, the protein bands are visualized using a detection method such as chemiluminescence.

Autosome dominant , autosome recessive, multifactorial, x linked recessive

Autosomal Dominant: Affected individuals are heterozygote. One allele is mutated, while other one is normal.

Autosomal recessive: Homozygous, when both alleles of the gene are mutated.

X-linked recessive: Mutated alleles are present on either of the sex chromosomes.

X-linked dominant: inheritance refers to genetic conditions associated with mutations in genes on the X chromosome. Families with an X-linked recessive disorder often have affected males, but rarely affected females, in each generation. For X-linked dominant diseases, however, a mutation in one copy of an X-linked gene will result in disease for both males and females.

Multifactorial refers to a trait or condition that is influenced by multiple factors, including both genetic and environmental factors. It means that the trait or condition is not solely determined by genetics, but also by other non-genetic factors such as lifestyle, diet, exposure to toxins, and other environmental influences.

Mendelian inheritance

Mendelian inheritance is the way traits pass from parents to offspring. Gregor Mendel found that traits are determined by "alleles" that segregate and assort independently. Some alleles are dominant, others recessive. This forms the basis of genetic inheritance.

Turner syndrome

Turner syndrome is a genetic disorder that affects females, typically resulting from a missing or partially missing X chromosome. It can lead to various physical and developmental features, including short stature, infertility, heart and kidney issues, and certain learning difficulties.

Names structure abnormalities

some structural abnormalities in chromosomes:

1. Deletion

2. Duplication

3. Translocation

4. Inversion

5. Ring Chromosome

6. Isochromosome

Giemsa stain

Geimsa stain is a type of stain used in microscopy to visualize and differentiate different types of cells, especially blood cells. It works by staining the DNA and RNA in the cells, allowing for better visualization of cellular structures and identification of specific cell types.

Short note epistasis

Epistasis is a genetic phenomenon where the effects of one gene are influenced by the presence of another gene. It can either mask the effects of another gene (recessive epistasis) or modify the expression of another gene (dominant epistasis). Epistasis plays a crucial role in shaping the complex interactions within genetic pathways and is an important consideration in understanding inheritance patterns and genetic traits.

Three name of diagnosis genetics diseases.

1-Cystic Fibrosis 2-Huntington's Disease 3-Down Syndrome

Trizol RNA

TriZol (or TRIzol) is a widely used reagent in molecular biology for the isolation of RNA, DNA, and protein from biological samples. It's used to extract RNA from cells and tissues, making it a crucial tool for various applications such as gene expression analysis, RNA sequencing.

Mitochondrial disease

Mitochondrial diseases are a group of genetic disorders that affect the mitochondria.

• Myoclonic epilepsy

• Mitochondrial recessive ataxia

• Leber hereditary optic neuropathy

• Sensory ataxia neuropathy

PCR uses /Applications of PCR

PCR has applications in DNA amplification,

1.Genetic testing

2 forensics, medical diagnostics,

3.Environmental studies

4.Evolutionary studies

5.food safety

6 .Archaeology, and anthropology

Genetic therapies.

Gene therapy is a procedure that involves identification, manipulation, and transference of genetic segments into a host to replace defective genes and to perform desired genetic activities.

VNTR and its applications

VNTR are short nucleotide sequences organized as a tandem repeats on genomes. Found on many chromosomes, Show variations in length between individuals.

Application:

1.DNA fingerprinting

2.genetic diversity

3. forensic mapping of genome

4. breeding pattern of wild and domesticated animals

5. paternity testing

6. Evolutionary research .

Structural variation of chromosomes

Structural variations of chromosomes refer to alterations in the normal structure of a chromosome, which can involve changes in the arrangement, size, or number of genetic material. These variations include deletions, duplications, inversions, and translocations, among others. They can lead to genetic disorders or contribute to genetic diversity in populations.

Point mutation, Transition mutation and transversion mutation

A point mutation or substitution is a genetic mutation where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. Point mutations have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the mutation.

If purine (A/G) is replaced by purines and pyrimidine (T/C) is replaced by pyrimidines it is called transitions mutation. If purine substitutes for pyrimidine or vice versa is called transversion

RAPD,ALFP difference

RAPD is Random Amplification of Polymorphic DNA'. It is a type of PCR, but the segments of DNA that are amplified are random.

AFLP is a PCR-based tool used in genetics research, DNA fingerprinting, and in the practice of genetic engineering. Developed in the early 1990s by Key gene, AFLP uses restriction enzymes to digest genomic DNA, followed by ligation of adaptors to the sticky ends of the restriction fragments.

Properties of mitochondrial DNA inherited

Mitochondrial DNA (mtDNA) is a type of genetic material found in the mitochondria of cells. It's inherited in a unique way compared to nuclear DNA. Here are some key properties of mitochondrial DNA inheritance:

1. Maternal Inheritance: Mitochondrial DNA (mtDNA) is inherited from the mother.

2. No Recombination: Unlike nuclear DNA, mtDNA doesn't recombine, staying stable.

3. Clonal Inheritance: mtDNA is passed down as a clonal unit.

4. Heteroplasmy/Homoplasmy: Variants in mtDNA (heteroplasmy) or just one type (homoplasmy) affect disease risk.

5. Threshold Effect: A certain % of faulty mtDNA is needed for symptoms.

6. Varied Symptoms: Similar mutations can lead to different disease severity.

7. Mitochondrial Disorders: mtDNA mutations cause energy-related disorders.

8. Evolutionary Insights: mtDNA helps study human evolution and population movement.

Differnce between purification of DNa through organic and inorganic methods.write down the step's of organic methods

The main difference between the inorganic method and the organic method of DNA extraction lies in the type of reagents used. In the inorganic method, substances like salts and alcohols are used to precipitate DNA, while the organic method involves the use of organic solvents like phenol and chloroform

Interphase and phases of cell cycle

Interphase is the stage in the cell cycle where the cell prepares for division.

It is divided into three phases: G1 (Gap 1), S (Synthesis), and G2 (Gap 2).

During interphase, the cell grows, carries out its normal functions, and replicates its DNA in the S phase. The cell cycle stages include interphase and mitosis (or meiosis in reproductive cells),

Continuous and discontinuous variation

Continuous variation refers to traits that can take on a range of values within a population, such as height or weight. These traits are influenced by multiple genes and environmental factors, resulting in a wide spectrum of possible values.

Discontinuous variation, on the other hand, refers to traits that can be categorized into distinct groups or phenotypes, such as blood type or eye color. These traits are usually controlled by a small number of genes with specific alleles, leading to clear-cut differences between categories

Autosomal recessive disease

An autosomal recessive disease is a genetic disorder caused by inheriting two copies of a mutated gene, one from each parent, who are typically carriers of the mutated gene. Examples of autosomal recessive diseases include cystic fibrosis, sickle cell anemia, and Tay-Sachs disease. These conditions often appear when an individual inherits two copies of the faulty gene, leading to the expression of the disease's symptoms. Carriers of a single copy of the mutated gene usually do not show symptoms but can pass the gene on to their offspring.

10 organism names with chromosomes

- Human: 46 chromosomes
- Mouse: 40 chromosomes
- Fruit Fly: 8 chromosomes
- Dog: 78 chromosomes
- Cat: 38 chromosomes
- Rice: 24 chromosomes
- Corn: 20 chromosomes
- chimpanzee :48 chromosomes
- Elephant: 56 chromosomes
- Tomato: 24 chromosomes

Genetic disease:

are conditions that are caused by changes or mutations in a person's genes. Some examples of genetic diseases include cystic fibrosis, sickle cell anemia, Huntington's disease, Down syndrome, and muscular dystrophy.

How protein are denatured and removed in DNA extraction

In DNA extraction, proteins are broken down using detergents and enzymes like proteinase K. Denaturing agents and heat unfold the proteins, and then separation methods like centrifugation or filtration remove denatured proteins, leaving the DNA for further processing.

In which form blood group exist

Blood groups exist in various forms due to different combinations of antigens and antibodies. The ABO blood group system involves four main blood types: A, B, AB, and O. These types are determined by the presence or absence of antigens A and B on the surface of red blood cells, along with antibodies against the missing antigens. The Rh factor, another important antigen, adds the positive (+) or negative (-) designation to the blood type (e.g., A+, B-, O-). This results in a variety of blood group combinations.

Ring and Philadelphia chromosome

A ring chromosome is a chromosome whose arms have fused together to form a ring. In humans it is denoted by small r and in drosophila it is denoted by capital R.

Philadelphia chromosome:- is a specific chromosomal abnormality that is related with chronic myelogenous leukemia(CML). Reciprocal translocation between chromosome 9 and chromosome 22 which is specifically designated.

Genetic and Environmental cause of variation

Variations in traits can be caused by both environmental and genetic factors. Environmental factors include things like diet, exposure to toxins, and physical activity, which can influence gene expression and affect traits. Genetic factors, on the other hand, involve variations in DNA sequences that are inherited from parents and can directly impact traits. It's important to note that in some cases variations are a result of a combination of both environmental and genetic influences.

Purines/ pyrimidine

The purines in DNA are adenine and guanine, the same as in RNA. The pyrimidines In DNA are cytosine and thymine; in RNA, they are cytosine and uracil. Purines are larger than pyrimidines because they have a two-ring structure while pyrimidines only have a single ring.

Nucleotide / Nucleosides

Nucleosides contain only sugar and a base whereas Nucleotides contain sugar, base and a phosphate group as well. A nucleotide is what occurs before RNA and DNA, while the nucleoside occurs before the nucleotide itself.

write 3 types of genetic disorder.

1. **Down Syndrome**: A chromosomal disorder caused by an extra copy of chromosome 21, leading to cognitive and physical developmental delays.

2. **Cystic Fibrosis**: A hereditary disorder affecting the lungs and digestive system, characterized by the production of thick and sticky mucus that can obstruct airways and lead to respiratory and digestive problems.

3. **Huntington's Disease**: An inherited neurological disorder that causes progressive degeneration of brain cells, leading to motor dysfunction, cognitive decline, and emotional disturbances.

Forward, reverse and neutral mutation

1. Forward Mutation: Alters one DNA base, possibly changing a protein's amino acid sequence.

2. Reverse Mutation: Mutated gene reverts back to its original sequence, restoring function.

3. Neutral Mutation: Doesn't impact an organism's traits or function noticeably.

Telophase 2 of meiosis

Telophase II is the final stage of meiosis II, which is the second round of cell division in meiosis. During Telophase II:

1. **Chromosomes**: Chromatids (sister chromatids) that were separated in Anaphase II reach opposite poles of the cell.

2. Nuclear Envelopes: New nuclear envelopes start forming around the separated chromatids, creating two new nuclei.

3. Cytokinesis: The cytoplasm divides, resulting in the formation of four distinct haploid daughter cells (gametes), each containing half the original number of chromosomes.

At the end of Telophase II, the process of meiosis is complete, and the resulting daughter cells are genetically varied due to the shuffling and separation of genetic material during meiosis I and II.

W.T.T

Explain trisomy and it's Types

Trisomy is a type of polysomy in which there are three copies of a particular chromosome,instead of the normal two. Trisomy is a type of aneuploidy.

Types

1. Trisomy 21 (Down syndrome), 2.Trisomy 18 (Edwards syndrome), and 3.Trisomy 13 (Patau syndrome).

Calculate the number of following for specific trait gene if 10 alleles for specific gene Genotype, homogyzous, heterozygous

If there are 10 different alleles for a specific gene and all of these alleles are equally represented in the population, here's how you can calculate the number of genotypes, homozygous genotypes, and heterozygous genotypes:

1. **Number of Genotypes**: The number of possible genotypes for a gene with 10 alleles can be calculated using the formula n(n+1)/2, where n is the number of alleles. So, for 10 alleles:

   Number of genotypes = 10 * (10 + 1) / 2 = 55 possible genotypes.

2. **Number of Homozygous** Genotypes: Homozygous genotypes occur when an individual has two identical alleles. Since there are 10 alleles, there are 10 possible homozygous genotypes.
3. **Number of Heterozygous** Genotypes: Heterozygous genotypes occur when an individual has two different alleles. Since there are 10 alleles and 55 total genotypes, the remaining 45 genotypes would be heterozygous.

So, in summary:

- Number of genotypes: 55

- Number of homozygous genotypes: 10

- Number of heterozygous genotypes: 45

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Please note that this assumes all alleles are equally represented and that there are no interactions or restrictions between the alleles. If the distribution of alleles is not equal or there are other factors at play, the calculations might differ.

Explain the experiment performed by T.H Morgan with and example

Morgan with an experiment proved that eye color trait is linked with sex of drosophila.

Experiment:

- Male flies were with white eyes(mutant).
- Female flies were with red eyes(wild type).
- Progeny f1 were with red eyes.
- All the flies produce during f1 generation were with red eyes.
- Progeny F2 showed 3:1 Ratio.(three red one white eyes.
- White eyes traits were appeared only in males fruit fly.
- All female and half male with red eyes.

Morgans Discovery

Transmission of X chromosomes correlates with inheritance of eye color traits. First solid evidence indicating that gene is associate with specific chromosomes.

Write about variation in autosomal dominant inheritance in reference of following terms.

Penetrance, expressivity, pleiotropy

Autosomal dominant inheritance can display variation due to factors such as penetrance, expressivity, and pleiotropy.

1. **Penetrance:** Penetrance refers to the percentage of individuals carrying a dominant allele who actually exhibit the associated trait. In some cases, the dominant allele might not always result in the observable trait, leading to incomplete penetrance. For instance, if a disease is caused by an autosomal dominant gene, individuals with the gene might not necessarily show symptoms.

2. **Expressivity**: Expressivity relates to the range of severity in which a trait is expressed among individuals who carry the same dominant allele. Even though a trait is dominant, its manifestation can vary in terms of intensity or features. For example, in a disorder caused by an autosomal dominant gene, one person might experience mild symptoms, while another might experience more severe symptoms.

3. **Pleiotropy**: Pleiotropy refers to a single gene influencing multiple traits or characteristics. In autosomal dominant inheritance, a single gene mutation can lead to a variety of effects in different parts of the body. For instance, a gene mutation responsible for a specific disorder might affect not only one organ system but also have secondary effects on other systems, resulting in a range of symptoms.

Reagent use in PCR.

Some of the key reagents used in a typical PCR reaction:

1. DNA Template: The starting DNA strand that contains the target sequence you want to amplify.

2. Primers: Short DNA sequences that define the start and end points of the target region to be amplified.

3. DNA Polymerase: The enzyme responsible for synthesizing a complementary DNA strand using the primers and nucleotides.

4. Nucleotides (dNTPs): The building blocks of DNA strands, containing adenine (A), thymine (T), cytosine (C), and guanine (G).

5. Buffer Solution: A solution that provides the optimal pH and salt concentration for the PCR reaction to occur.

These reagents work together in the PCR process to amplify the target DNA region.

Precautions during extraction of RNA

- Use RNase-free equipment and reagents.
- Wear gloves to prevent skin contamination.
- Work in a clean area to avoid contamination.
- Consider RNA stabilizing agents for storage.
- Minimize mechanical shearing of RNA.
- Avoid exposing RNA to high temperatures.
- Homogenize carefully for even RNA distribution.
- Process samples quickly to prevent degradation.
- Prevent cross-contamination between samples.
- Follow established RNA extraction protocols.
- Perform quality checks after extraction.
- Keep detailed documentation of the proc

Diagnostic test for genetic disorder

A diagnostic test for a genetic disorder usually involves analyzing a person's DNA to detect specific mutations or variations associated with the disorder. These tests can include techniques like PCR, DNA sequencing, and microarray analysis. Keep in mind that specific tests depend on the suspected disorder. It's important to consult a medical professional for accurate guidance and testing recommendations.

HGCN rule for substitution, range of residue duplication deletion insertion inversion

The HGCN rule, or the Human Genome Variation Society nomenclature, provides guidelines for describing genetic variations in DNA, RNA, and protein sequences. The HGCN rule includes specific conventions for representing different types of variations, such as substitutions, insertions, deletions, and rearrangements, in a standardized format. This allows researchers and clinicians to accurately communicate and interpret genetic variations across different studies and databases. The HGCN rule plays a crucial role in facilitating the understanding and sharing of genetic information in the scientific community.

- "c." for a coding DNA sequence (like c.76A>T)
- "g." for a genomic sequence (like g.476A>T)
- "m." for a mitochondrial sequence (like m.8993T>C)
- “r.” for an RNA sequence (like r.76a>u)
- ”p.” for a protein sequence (like p.Lys76Asn)

Polygenic traits with example

Polygenic traits are traits that are influenced by multiple genes, each contributing a small effect to the overall phenotype. These traits typically show a continuous variation rather than distinct categories. An example of a polygenic trait is human height. Height is determined by the combined effects of multiple genes, making it a continuous distribution of heights rather than just a few discrete options.

Difference between telophase I and telophase II (3)

Telophase I:

- Occurs after first meiotic division (Meiosis I).

- Daughter cells are haploid (n).

- Homologous chromosomes separate.

- Genetic diversity introduced due to crossing over.

- Results in two haploid daughter cells.

Telophase II:

- Occurs after second meiotic division (Meiosis II).

- Daughter cells are also haploid (n).

- Sister chromatids separate.

- Genetic diversity maintained from Telophase I.

- Results in four haploid daughter cells (gametes).

W.T.T

Clinical signs of down syndrome (5)

some clinical signs often associated with Down syndrome:

1. Distinct facial features
2. Hypotonia (low muscle tone)
3. Developmental delays
4. Cognitive impairment
5. Heart defects
6. Gastrointestinal issues
7. Hearing and vision problems
8. Thyroid issues
9. Short stature
10. Speech and language difficulties

Loss of function and gain of function (3)

Loss of Function:

Loss of function refers to genetic mutations that lead to a decrease or elimination of a gene's or protein's normal activity. This can result in impaired cellular processes, often causing genetic disorders due to the disrupted function of essential components.

Gain of Function:

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Gain of function involves genetic mutations that cause a gene's or protein's activity to increase or acquire new functions. This can lead to enhanced cellular activity, sometimes contributing to disease development or providing an advantage in evolution.

Effects of deletion (3)

Deletion is the removal of a segment of DNA from a chromosome, resulting in the loss of genetic material.

Effects of Deletion:

1. Gene Loss: Can lead to the absence of a functional protein.

2. Haplo insufficiency: Reduction in gene dosage affecting normal cellular function.

3. Dominant Negative Effects: Non-functional protein interferes with the function of the normal protein.

4. Chromosomal Imbalance: Disruption of genetic material balance.

5. Altered Regulatory Regions: Impact on gene expression control.

Long term impacts of genetics .

Long-Term Impacts of Genetics:

- Healthcare: Personalized medicine, disease prediction, and drug development.

- Evolution: Natural selection, speciation, and genetic diversity.

- Agriculture: Genetically modified crops, improved breeding programs.

- Forensics: DNA identification, tracing human migration.

- Ethics: Genetic privacy, ethical concerns in genetic engineering.

- Scientific Advancement: Fundamental research, technological innovation.

- Conservation: Biodiversity protection, endangered species conservation.

- Understanding Life: Molecular biology, uncovering life's mechanisms.

why we study polymorphism in organism?

Essential to study inheritance patterns. Map phenotypes and anchor genes to the genetic map by co-segregation analysis. Determine change in function: resistant/sensitive populations genetically determined variability among humans is due to a difference in 0.1% of the genomic sequence.

write down the method which convert euchromatin into heterochromatin

Histone deacetylation catalyzed by histone deacetylases (HDACs) removes acetyl groups from histone proteins. This leads to tighter histone-DNA interactions, causing chromatin compaction and converting active euchromatin to inactive heterochromatin. This process regulates gene expression and is crucial for various biological processes.

Write about the non.disjunction

Non-disjunction is a genetic event that can occur during cell division, specifically during the separation of chromosomes. It can be caused by various factors, including genetic abnormalities, environmental factors, and maternal age. Non-disjunction can lead to an abnormal number of chromosomes in the resulting cells, which can cause genetic disorders such as Down syndrome.

Define heterozygous dominant trait with example.

Heterozygous dominant refers to a genetic condition where an individual possesses two different alleles for a particular gene, and the dominant allele's trait is expressed in the phenotype.

Example: Tongue Rolling

Dominant Trait: Ability to roll the tongue (T)

Recessive Trait: Inability to roll the tongue (t)

Heterozygous Dominant: (Tt) individual can roll their tongue.

Explain mendelian and non mendelian inheritance with example.

Mendelian genetics refers to the laws of inheritance discovered by Gregor Mendel in the 19th century. These laws describe the patterns of inheritance of traits that are controlled by a single gene, such as the inheritance of flower color in pea plants. Mendelian genetics involves the inheritance of traits that are determined by a single gene locus and are either dominant or recessive.

Non-Mendelian genetics refers to patterns of inheritance that do not follow the simple dominant/recessive pattern described by Mendel's laws. These patterns of inheritance can be due to a variety of factors, such as incomplete dominance, co-dominance, multiple alleles, or sex-linked inheritance. In incomplete dominance, neither allele is dominant, nor is the phenotype of the heterozygote a blend of the two parental phenotypes. In co-dominance, both alleles are expressed in the phenotype of the heterozygote. Multiple alleles refer to the presence of more than two alleles for a given gene, while sex-linked inheritance refers to the inheritance of genes that are located on the sex chromosomes.

Explain the haploid- diploid sex determination system

The haploid-diploid sex determination system is a type of sex determination found in certain organisms, notably some insects like bees and ants, and some plants. In this system, the sex of an individual is determined by the number of sets of chromosomes it possesses.

Haploid (n):Haploid individuals have one set of chromosomes. They develop from unfertilized eggs (parthenogenesis) and are usually males.

Diploid (2n):Diploid individuals have two sets of chromosomes. They develop from fertilized eggs and can be either males or females.

In this system, males are haploid because they develop from unfertilized eggs and only inherit half the number of chromosomes. Females are diploid because they develop from fertilized eggs and inherit the full complement of chromosomes.

This system is quite different from the familiar XX-XY sex determination in humans, where females have two X chromosomes (XX) and males have one X and one Y chromosome (XY).

This haploid-diploid system can result in some unique behaviors and social structures, particularly in insects like bees and ants, where females are more closely related to their sisters than to their own potential offspring.

Prophase 2 of meiosis?

During Prophase II of meiosis II:

- Chromosomes condense.

- Nuclear envelope dissolves.

- Spindle fibers form.

- Chromosomes align in the center of the cell.

how mutation alter the gene product?

Mutations alter gene products by:

- Changing amino acid sequence.

- Introducing premature stop codons.

- Causing frameshifts (insertions/deletions).

- Disrupting splice sites and mRNA splicing.

- Expanding repeat sequences.

- Occasionally having no impact (silent mutations).

- Occasionally causing gain-of-function effects.

Types of chromosome on the base of centromere location

Based on the location of the centromere, chromosomes can be classified into four types:

W.T.T

1.metacentric

2.submetacentric

3. acrocentric

4. telocentric.

Drawbacks of mendel law of independent assortment

Drawbacks of Mendel's Law of Independent Assortment:

- Genetic linkage violates independence.

- Exceptions due to incomplete dominance, etc.

- Complex traits involve multiple alleles and genes.

- Crossing over generates new allele combinations.

- Doesn't cover inheritance of sex-linked traits.

- Ignores gene interactions like epistasis.

- Limited to model organism's traits.

- Ignores environmental influences.

- Incomplete penetrance and variable expressivity not considered.

Difference between prokaryotic and eukaryotic chromosome

Prokaryotic chromosomes are single, circular structures located in the nucleoid region. They are generally less compact compared to eukaryotic chromosomes due to the absence of histones. Prokaryotic cells typically contain one chromosome each, with a relatively simple organization and limited non-coding DNA. Replication of prokaryotic chromosomes begins at a single origin, and cell division occurs through binary fission. These chromosomes are generally smaller in size, often carry clustered genes in polycistronic operons, and lack introns.

On the other hand, **eukaryotic chromosome**s are multiple, linear structures found within the membrane-bound nucleus. They are highly compacted with the help of histones. Eukaryotic cells have multiple chromosomes, each with a more complex organization that includes significant non-coding DNA regions. Replication of eukaryotic chromosomes involves multiple origins. Cell division occurs through mitosis in somatic cells and meiosis in germ cells. Eukaryotic chromosomes are larger, commonly carry single-cistronic genes, and exhibit a lesser occurrence of operons. They often contain introns, necessitating splicing for proper gene expression.

Downstream genetic mapping.

Downstream genetic mapping refers to the process of analyzing genetic data to identify and map specific genes or genetic markers that are associated with a particular trait or disease. This involves conducting statistical analyses, such as genome-wide association studies (GWAS), to identify regions of the genome that are linked to the trait of interest. This information can then be used to further investigate the underlying genetic mechanisms and potential therapeutic targets.

Define leading strand, lagging strand and sister chromatids??

- The leading strand is the strand of DNA that is synthesized continuously during DNA replication.It is synthesized in the 5' to 3' direction, which is the same direction as the movement of the replication fork.
- The lagging strand is the strand of DNA that is synthesized discontinuously during DNA replication. It is synthesized in short fragments called Okazaki fragments, which are later joined together to form a continuous strand.
- Sister chromatids are two identical copies of a chromosome that are produced during DNA replication. They are held together by a structure called the centromere. Sister chromatids are separated during cell division and are distributed to the daughter cells.

Intergenic and Intragenic Mutations

Intergenic mutations occur in the non-coding regions of DNA between genes, while intragenic mutations occur within a gene's coding or non-coding regions. Intergenic mutations often have less direct impact on gene function, while intragenic mutations can affect a gene's structure or function, potentially leading to genetic disorders.

How we can identify chromosome?

We can identify chromosomes through various techniques, such as karyotyping, fluorescent in situ hybridization (FISH), or chromosomal microarray analysis (CMA). These methods allow us to visualize and analyze the structure and number of chromosomes in a cell.

----------SPECIAL REGARDS----------

♡ ATTARIYA'S WORK TOGETHER TEAM ♡

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